

Homework 2

Numerical methods, Practice

The below problems are optional homework exercises related to the topics of the second half of the semester. These may be handed in as announced by the practice teacher (usually uploaded to the Canvas system) with the deadline of the second test. One may gain at most 5 extra points by solving some of these problems. The complete solution of a problem is worth 1 point.

Problems

1. Find an iteration method to evaluate the expression

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$$

Prove its convergence and give the order of convergence.

2. Consider the iteration $x_{n+1} = \sqrt{\cos(x_n)}$. Study its convergence. Find an interval for x_0 such that the fixed point iteration converges.
3. Apply Newton's method to approximate the solution of the equation $x - e^{-x} = 0$. Show that the method converges for all $0 < x_0 < 1$.
4. Show that the function y which fulfills the equation

$$\begin{vmatrix} y(x) & 1 & x & x^2 & \cdots & x^n \\ f(x_0) & 1 & x_0 & x_0^2 & \cdots & x_0^n \\ f(x_1) & 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ f(x_n) & 1 & x_n & x_n^2 & \cdots & x_n^n \end{vmatrix} = 0$$

is equal to the interpolating polynomial L_n of the function f with respect to the nodes $x_0, x_1, \dots, x_n$.

5. Show that on the nodes $x_0, x_1, \dots, x_n$ ($n \geq 2$)

$$\sum_{j=0}^n (x - x_j)^k \ell_j(x) = \begin{cases} 0 & \text{if } 1 \leq k \leq n \\ 1 & \text{if } k = 0 \end{cases}$$

where $\ell_j(x)$ denotes the fundamental polynomial of Lagrange interpolation with respect to x_j .

6. Show that on the nodes $x_0, x_1, \dots, x_n$

$$\sum_{j=0}^n \frac{x_j^k}{\omega'_n(x_j)} = \begin{cases} 0 & \text{if } 0 \leq k \leq n-1 \\ 1 & \text{if } k = n \end{cases}$$

where $\omega_n(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$.

7. Show that on the nodes $x_0, x_1, \dots, x_n$ with ℓ_j and ω_n as above the following equality holds:

$$x^{n+1} - \sum_{j=0}^n x_j^{n+1} \ell_j(x) = \omega_n(x).$$

8. It is known that

$$\sum_{k=0}^n k^3 = P_4(n)$$

where P_4 is a polynomial of degree 4. Find P_4 using interpolation.

9. Consider the quadrature formula

$$\int_0^1 f(x) \sqrt{x} \, dx \approx \sum_{k=1}^n A_k^{(n)} \cdot f\left(\frac{k}{n}\right)$$

which is exact for polynomials of degree at most n . Find the coefficients of the formula for $n = 3$ and apply it to approximate the integral

$$\int_0^1 \sqrt{x} \cos x \, dx.$$

10. Consider the quadrature formula

$$\int_0^1 f(x) \, dx \approx A_0 \cdot f(x_0) + A_1 \cdot f(x_1).$$

Find A_0 , A_1 and x_0 , x_1 for the highest possible degree of precision.